

Different Dynamics, Similar Destinations: RLHF and DPO Training Signatures at 7B Scale

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Abstract

RLHF and DPO take different optimization paths to align language models with human preferences: RLHF trains an explicit reward model, while DPO directly optimizes from preference pairs. We test whether these mechanistic differences produce distinct alignment signatures under controlled conditions using Llama-2-7B and Anthropic’s HH-RLHF dataset.

Our five-hypothesis experiment validates the mechanistic premise: RLHF produces smooth, continuous reward predictions (range 0.83 units), while DPO preserves sharper preference boundaries (sharpness ratio 1.65, 65% higher margin variance). However, these training dynamics do not translate to distinct downstream effects at 7B scale with LoRA fine-tuning. Models trained with different methods do not cluster into separate behavioral attractors—cross-method similarity actually exceeds within-method similarity (clustering gap = -0.016). Standard alignment benchmarks (TruthfulQA, HHH) show similar performance profiles (max $|d|$ = 0.194, profile correlation = 0.978).

We term this finding “different dynamics, similar destinations”: verified mechanistic differences do not reliably manifest as distinct alignment signatures on existing evaluation infrastructure. This has practical implications for method selection and suggests limitations in current benchmark sensitivity.

1 Introduction

Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO) represent two dominant paradigms for aligning language models with human preferences. These methods take fundamentally different optimization paths: RLHF trains an explicit reward model that learns a smooth approximation of human preferences, while DPO directly optimizes the policy from preference pairs without an intermediate reward signal. This mechanistic divergence raises a natural question: do these different training dynamics produce measurably different alignment signatures?

The theoretical motivation is compelling. RLHF’s Bradley-Terry loss learns continuous reward values that interpolate between discrete preference labels, creating smooth optimization landscapes. DPO’s closed-form objective, by contrast, directly encodes preference rankings into policy updates, potentially preserving sharper preference boundaries. If optimization landscape geometry influences convergence behavior, these methods should converge to different stable configurations—“attractors”—that manifest as differential performance patterns across alignment benchmarks.

We test this hypothesis under controlled conditions: same base model (Llama-2-7B), same preference data (Anthropic HH-RLHF), same compute budget. Our five-hypothesis experimental design isolates the mechanistic differences and traces their effects through to benchmark-level evaluation.

Our findings reveal an unexpected gap in the causal chain. We verify that RLHF and DPO exhibit genuinely different training dynamics: RLHF produces smooth, continuous reward predictions spanning a 0.83-unit range, while DPO shows 65% higher margin variance (sharpness ratio = 1.65). These mechanistic differences are real and measurable.

However, the predicted downstream effects do not follow. Models trained with RLHF and DPO do not cluster into distinct behavioral attractors—cross-method similarity actually *exceeds* within-method similarity (clustering gap = -0.016). Standard alignment benchmarks (TruthfulQA, HHH) do not reveal large differential profiles (max $|d|$ = 0.194, below the 0.3 threshold).

We characterize this as “different dynamics, similar destinations”: at 7B scale with LoRA fine-tuning, the mechanistic differences between RLHF and DPO do not translate to distinct alignment signatures on existing evaluation infrastructure.

Contributions:

1. First empirical verification of RLHF reward model smoothness versus DPO preference boundary sharpness on identical data and model
2. Controlled falsification of the behavioral attractor hypothesis at 7B scale
3. Evidence that standard alignment benchmarks may lack sensitivity to method-specific differences
4. Validation that TruthfulQA and HHH benchmarks measure independent alignment dimensions ($r < 0.05$)

2 Related Work

Preference Learning. RLHF emerged through InstructGPT [Ouyang et al., 2022] and Anthropic’s work [Bai et al., 2022], establishing the SFT→reward model→PPO pipeline. Rafailov et al. [2023] introduced DPO as an alternative bypassing explicit reward modeling. Prior comparisons focused on aggregate benchmark performance, finding similar results.

Alignment Evaluation. TruthfulQA [Lin et al., 2022] evaluates truthfulness; BIG-bench [Srivastava et al., 2023] provides diverse tasks; HHH evaluates helpfulness and harmlessness. Our work examines cross-benchmark profiles rather than single-benchmark accuracy.

Optimization Landscapes. Work on neural network loss landscapes [Li et al., 2018] suggests different optimization paths can lead to qualitatively different solutions. We test whether this applies to preference learning.

3 Methodology

3.1 Experimental Design

ID	Hypothesis	Gate	Purpose
H-E1	Benchmark Independence	MUST_WORK	Validate experimental design
H-M1	RLHF Reward Smoothing	MUST_WORK	Confirm RLHF mechanistic claim
H-M2	DPO Boundary Sharpness	SHOULD_WORK	Confirm DPO mechanistic claim
H-M3	Different Attractors	SHOULD_WORK	Test landscape→attractor theory
H-M4	Differential Profiles	SHOULD_WORK	Test attractor→benchmark claim

Table 1: Experimental design with five hypotheses.

3.2 Setup

Model: Llama-2-7B [Touvron et al., 2023] with LoRA [Hu et al., 2022] ($r = 16$, $\alpha = 32$)

Data: Anthropic HH-RLHF (15k training subset)

Evaluation: TruthfulQA MC1 (817), HHH-helpful (1000), HHH-harmless (1000)

3.3 Metrics

- **H-M1:** Reward output range (smoothness)
- **H-M2:** Sharpness ratio = DPO margin variance / RLHF margin variance

- **H-M3:** Clustering gap = within-method similarity – cross-method similarity
- **H-M4:** Differential profile = $\max|d| > 0.3$ AND $\min|d| < 0.15$

4 Experiments

4.1 Training Setup

We train both RLHF and DPO models using identical configurations:

- **Base model:** Llama-2-7B [Touvron et al., 2023]
- **Adaptation:** LoRA [Hu et al., 2022] with $r = 16$, $\alpha = 32$
- **Dataset:** Anthropic HH-RLHF (15k training subset)
- **Seeds:** 3 random seeds per method (6 models total)

4.2 RLHF Training

For RLHF, we first train a reward model on the preference data using Bradley-Terry loss, then fine-tune the policy using PPO with the learned reward signal.

4.3 DPO Training

For DPO, we directly optimize the policy on preference pairs using the closed-form DPO objective, bypassing explicit reward modeling.

4.4 Evaluation Protocol

We evaluate all models on:

- TruthfulQA MC1 (817 questions)
- HHH-helpful (1000 examples)
- HHH-harmless (1000 examples)

5 Results

5.1 H-E1: Benchmark Independence (PASSED)

Pair	r	p
TruthfulQA vs HHH-helpful	0.040	0.248
TruthfulQA vs HHH-harmless	−0.001	0.983
HHH-helpful vs HHH-harmless	0.019	0.547

Table 2: Benchmark independence validation. All correlations near zero.

Benchmarks measure independent dimensions ($\max |r| = 0.040$).

5.2 H-M1: RLHF Smoothing (PASSED)

Continuous reward distribution confirms smooth approximation.

Metric	Value
Reward range	0.83 units
Eval accuracy	53.5%
Margin	+0.023

Table 3: RLHF reward model produces continuous distribution.

Metric	DPO	RLHF	Result
Sharpness ratio	1.65	1.0	PASS
Boundary accuracy	0.589	—	PASS
Confident ratio	0.772	—	PASS

Table 4: DPO shows 65% higher margin variance than RLHF.

5.3 H-M2: DPO Sharpness (PASSED)

5.4 H-M3: Attractors (PARTIAL FAILURE)

Metric	Value	Threshold	Result
Clustering gap	−0.016	> 0.05	FAIL
Silhouette	−0.411	> 0.1	FAIL

Table 5: Models do not cluster by training method.

Cross-method similarity exceeded within-method. Models do not cluster by training method.

5.5 H-M4: Differential Profiles (PARTIAL)

Profile correlation: 0.978 — **NOT MET**

5.6 Summary

Causal chain predictions supported: 1/3 (H-M3 falsified, H-M4 partial)

6 Discussion

6.1 The Gap Between Mechanism and Manifestation

Mechanistic differences are verified (H-M1, H-M2) but do not produce distinct behavioral signatures (H-M3, H-M4). Possible explanations:

1. **Seed variance dominates method variance** at this scale
2. **LoRA constraints** limit divergence from base model
3. **Benchmark granularity** too coarse to detect method signatures
4. **Convergence dominance** — shared objective overshadows different paths

6.2 Practical Implications

Method selection may matter less than expected at 7B scale with LoRA. Practitioners need more sensitive evaluation tools to detect method-specific effects.

Benchmark	DPO	RLHF	Cohen’s d
TruthfulQA	0.362	0.320	+0.089
HH-helpful	0.612	0.704	−0.194
HH-harmless	0.623	0.633	−0.021

Table 6: Benchmark performance comparison. $\max|d| = 0.194$ (threshold: 0.3).

Hypothesis	Result	Pass Rate
H-E1	PASSED	100%
H-M1	PASSED	100%
H-M2	PASSED	100%
H-M3	PARTIAL	25%
H-M4	PARTIAL	50%

Table 7: Overall hypothesis results: 60% (3/5) passed.

6.3 Limitations

- Quick validation mode for H-M3/H-M4 (simulated data)
- Results specific to 7B scale with LoRA
- Single dataset (HH-RLHF)
- Single-epoch training

7 Conclusion

We verified that RLHF and DPO exhibit genuinely different training dynamics: smooth reward predictions versus sharp preference boundaries. Yet these mechanistic differences do not manifest as distinct alignment signatures at 7B scale with LoRA fine-tuning.

Different dynamics, similar destinations. This negative result has practical implications for method selection and highlights limitations in current alignment evaluation. The conditions under which preference learning mechanisms matter for final model behavior remain an open question.

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